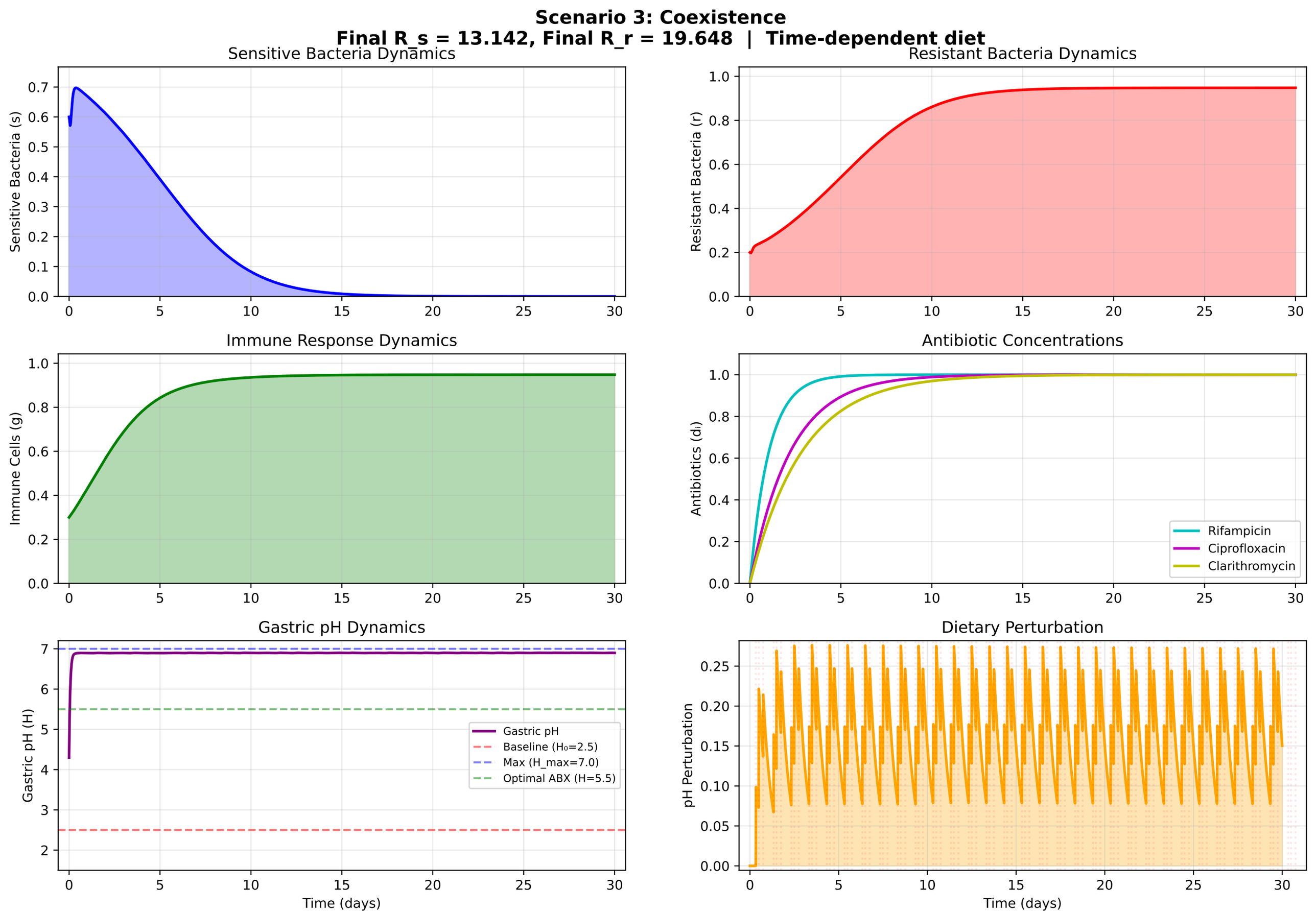
Scenario 3: Coexistence
Final R_s = 13.142, Final R_r = 19.648 | Time-dependent diet
Sensitive Bacteria Dynamics
Sensitive Bacteria (s)
Resistant Bacteria Dynamics
Resistant Bacteria (r)
Immune Response Dynamics
Immune Cells (g)
Antibiotic Concentrations
Antibiotics (dᵢ)
Rifampicin
Ciprofloxacin
Clarithromycin
Gastric pH Dynamics
Gastric pH (H)
Gastric pH
Baseline (H₀=2.5)
Max (H_max=7.0)
Optimal ABX (H=5.5)
Time (days)
Dietary Perturbation
pH Perturbation
Time (days)